

CPP (Fundamentals)

1) Basic I/O Setup

```
ios::sync_with_stdio(false);  
cin.tie(nullptr);
```

↳ disable stream synchronizations

2) Basic Input types

float/double

```
double x; cin >> x; cout << fixed << setprecision(6)  
        << x;
```

shows in non scientific
format

string (no spaces)

```
string s; cin >> s; cout << s;
```

line with spaces

```
string lined; getline(cin, lined); cout << lined;
```

fixed array

```
int n;
```

```
cin >> n;
```

```
int arr[n];
```

```
for (int i = 0; i < n; i++) cin >> arr[i];
```

2D Input/ Matrix

```
int n, m;
```

```
cin >> n >> m;
```

```
vector<vector<int>> mat(n, vector<int>(m));
```

```
for (int i = 0; i < n; i++) {
```



```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    string line;
```

```
    getline(cin, line);  $\leadsto$  1,2,3,4,5,6
```

```
    vector<int> nums;
```

```
    stringstream ss(line);  $\leadsto$  create fake stream but string can  
                                do both input/output
```

```
    string token;
```

```
    while (get(ss, token, ',')) {
```

```
        nums.push_back(stoi(token));
```

```
    }
```

```
}
```

5) Type Def

string $\rightarrow$ int stoi(s)

string $\rightarrow$ long long stoll(s)

string $\rightarrow$ float stof(s)

string $\rightarrow$ double stod(s)

int $\rightarrow$ string to_string(x)

6) Map $\hookrightarrow$ done using transform

```
vector<int> nums = {1,2,3,4,5};
```

```
vector<int> doubled(nums.size());
```

```
transform(nums.begin(), nums.end(), doubled.begin(),
```

```
    [](int x) {
```

```
        return x * 2;
```

```
    });
```


7) filter

↳ done using copy_if()

```
vector<int> nums = {1, 2, 3, 4, 5, 6}
```

```
vector<int> evens;
```

```
copy_if(nums.begin(), nums.end(), back_inserter(evens), [](int x){  
    return x % 2 == 0;  
});
```

Deque (Modifier)

1) clear

```
int main() {  
    deque<int> d = {1, 2, 3};  
    d.clear();  
    cout << d.size(); // 0  
}
```

2) insert

```
deque<int> d = {1, 3, 4};  
d.insert(d.begin() + 1, 2); // {1, 2, 3, 4}
```

3) insert_range() [C++23]

```
deque<int> d = {1, 2};  
vector<int> v = {3, 4, 5};  
d.insert_range(d.end(), v); // [1, 2, 3, 4, 5]
```

4) erase()

```
deque<int> d = {1, 2, 3, 4};  
d.erase(d.begin() + 1); // remove 2 → [1, 3, 4]  
d.erase(d.begin(), d.begin() + 2); // remove first two → [4]
```


5) push_back()

```
deque<int> d = {1, 2};
```

```
d.push_back(3); // [1, 2, 3]
```

6) Append_range [C++23]

```
deque<int> d = {1};
```

```
vector<int> v = {2, 3, 4};
```

```
d.append_range(v); // [1, 2, 3, 4];
```

7) Pop_back()

```
deque<int> d = {1, 2, 3};
```

```
d.pop_back(); // [1, 2]
```

8) Push_front

```
deque<int> d = {2, 3};
```

```
d.push_front(1); // [1, 2, 3]
```

9) Prepend_range

```
deque<int> d = {3, 4};
```

```
vector<int> v = {1, 2};
```

```
d.prepend_range(v); // [1, 2, 3, 4]
```

10) push_front()

```
deque<int> d = {2, 3};
```

```
d.push_front(1); // [1, 2, 3];
```

11) Pop_front

```
deque<int> d = {1, 2, 3};
```

```
d.pop_front(); // [2, 3];
```

12) resize()

```
deque<int> d = {1, 2, 3};
```

```
d.resize(5, 0); // [1, 2, 3, 0, 0]
```


13) swap()

```
deque<int> a = {1, 2};
```

```
deque<int> b = {3, 4};
```

```
a.swap(b); // a = {3, 4}, b = {1, 2}
```

11) Set

↳ store unique element (no duplicate)

↳ keep them sorted automatically

↳ uses a balanced BST (RB Tree) → $O(\log n)$ insertion

```
#include <iostream>
```

```
#include <set>
```

```
using namespace std;
```

```
int main() {
```

```
    set<int> s = {3, 1, 4};
```

```
    s.insert(2);
```

```
    for (int x : s) cout << x << " "; // Output - 1 2 3 4
```

```
}
```

Iterating over a Set

1) Range-based for loop

```
for (int x : s) cout << x << " ";
```

2) Reverse Iterator

```
for (auto it = s.rbegin(); it != s.rend(); it++) {  
    cout << *it << " ";
```

```
}
```

1) Count()

↳ returns 1 if the element exist, otherwise 0.

```
cout << s.count(20) << endl; → return 1 (exists)
```

```
cout << s.count(25) << endl; → return 0 (does not exist)
```


↳ check quickly if an element ($O(\log n)$).

2) find()

↳ find an element and return an iterator to it (or s.end()) if not found.

```
auto it = s.find(30);
```

```
if (it != s.end()) {
```

```
    cout << "found: " << *it << endl;
```

```
} else {
```

```
    cout << "Not found" << endl;
```

3) contain()

↳ simpler version of find()

```
if (s.contains(40)) {
```

```
    cout << "Set contains 40" << endl;
```

```
} else {
```

```
    cout << "No 40" << endl;
```

4) equal_range()

↳ return a pair of iterators - lower bound and upper bound for elements equal to a given key.

```
auto [low, hi] = s.equal_range(2);
```

5) lower_bound(x)

6) upper_bound(x)

1v) Map

↳ Map is a sorted associative container that contains key-value pairs with unique keys.

1) clear

↳ Remove all elements from the map, leaving it with a size of 0.

```
scores.clear();
```

2) insert

↳ insert a new key-value pair. if the key already exist insert does nothing. it returns a `std::pair` containing an iterator to the element (either the newly inserted one or the existing one) and a `bool` which is `true` if the insertion took place, & `false` otherwise.

```
int main() {  
    std::map<std::string, int> scores = {"Alice", 95};  
    print_map(scores);  
  
    // Insert a new element - succeeds  
    auto result1 = scores.insert({"Bob", 88});  
    std::cout << "Inserting 'Bob': success = " << std::boolalpha << result1.second << std::endl;  
    print_map(scores);  
  
    // Try to insert an existing element - fails  
    auto result2 = scores.insert({"Alice", 100});  
    std::cout << "Inserting 'Alice' again: success = " << result2.second << std::endl;  
    print_map(scores); // Alice's score remains 95  
    return 0;  
}
```

```
{ 'Alice': 95 }  
Inserting 'Bob': success = true  
{ 'Alice': 95 'Bob': 88 }  
Inserting 'Alice' again: success = false  
{ 'Alice': 95 'Bob': 88 }
```

3) insert_range

↳ insert a range of element from another container
more efficient than inserting element one by one in a loop.


```
int main() {
    std::map<std::string, int> scores = {"Alice", 95};
    std::vector<std::pair<std::string, int>> new_scores = {"Charlie", 92}, {"David", 78};

    std::cout << "Before insert_range: ";
    print_map(scores);

    scores.insert_range(new_scores);

    std::cout << "After insert_range: ";
    print_map(scores);
    return 0;
}
```

Before insert_range: { 'Alice': 95 }

After insert_range: { 'Alice': 95 'Charlie': 92 'David': 78 }

insert_or_assign(C++17)

↳ This is a very useful function. It inserts a key-value pair if the key does not exist, it updates the value associated with that key. This avoids the common "find-then-update or insert" pattern.

```
int main() {
    std::map<std::string, int> scores = {"Alice", 95};
    print_map(scores);

    // Insert a new element
    scores.insert_or_assign("Bob", 88);
    std::cout << "After inserting 'Bob': ";
    print_map(scores);

    // Update an existing element
    scores.insert_or_assign("Alice", 100);
    std::cout << "After assigning 'Alice': ";
    print_map(scores);
    return 0;
}
```

{ 'Alice': 95 }

After inserting 'Bob': { 'Alice': 95 'Bob': 88 }

After assigning 'Alice': { 'Alice': 100 'Bob': 88 }

erase

↳ Remove element from the map. It can be used in three ways:

- 1) By key → Removes the element with the specified key.
- 2) By iterator → Removes the element at the iterator's position.
- 3) By range → Removes all elements in the range [first, last).


```
int main() {
    std::map<std::string, int> scores = {"Alice", 95}, {"Bob", 88}, {"Charlie", 92}, {"David", 78};
    print_map(scores);

    // 1. Erase by key
    scores.erase("Bob");
    std::cout << "After erasing 'Bob': ";
    print_map(scores);

    // 2. Erase by iterator
    auto it = scores.find("David");
    if (it != scores.end()) {
        scores.erase(it);
    }
    std::cout << "After erasing 'David': ";
    print_map(scores);
    return 0;
}
```

{ 'Alice': 95 'Bob': 88 'Charlie': 92 'David': 78 }

After erasing 'Bob': { 'Alice': 95 'Charlie': 92 'David': 78 }

After erasing 'David': { 'Alice': 95 'Charlie': 92 }

Swap

↳ Exchange the content of two maps. This is an extremely fast operation ($O(1)$) because it just swap internal pointer, not the actual element.

```
int main() {
    std::map<std::string, int> map1 = {"A", 1}, {"B", 2};
    std::map<std::string, int> map2 = {"X", 10}, {"Y", 20};

    std::cout << "Map1 before swap: "; print_map(map1);
    std::cout << "Map2 before swap: "; print_map(map2);

    map1.swap(map2);

    std::cout << "Map1 after swap: "; print_map(map1);
    std::cout << "Map2 after swap: "; print_map(map2);
    return 0;
}
```

Map1 before swap: { 'A': 1 'B': 2 }

Map2 before swap: { 'X': 10 'Y': 20 }

Map1 after swap: { 'X': 10 'Y': 20 }

Map2 after swap: { 'A': 1 'B': 2 }

extract(C++17)

↳ Removes a node (key value) Pair from the map without destroying it. It returns a 'node handle' that own the element, This can be used to re-insert into the same map or another

compatible map, which is efficient.

```
int main(){
    std::map<std::string, int> scores = {"Alice", 95}, {"Bob", 88};

    // Extract the node for "Bob"
    auto node_handle = scores.extract("Bob");

    std::cout << "Scores after extracting 'Bob': ";
    print_map(scores);

    // We can modify the value in the extracted node
    node_handle.mapped() = 90;

    // Insert the modified node back into the map
    scores.insert(std::move(node_handle));

    std::cout << "Scores after inserting node back: ";
    print_map(scores);
    return 0;
}
```

→ move is required as node_handle is movable
but can't be copied.

Scores after extracting 'Bob': { 'Alice': 95 }

Scores after inserting node back: { 'Alice': 95 'Bob': 90 }